Mid-Sem Exam Solutions

Ques 1. Consider the circuit shown in Fig. 1. Assume base-to-emitter voltage V_{BE}=0.8 V and common emiiter current gain β=100.
(a) Find I_B, I_C, I_E and V_{CE}
(b) Identify the region of operation

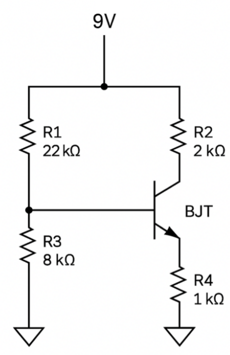

Fig. 1

Soln. (a)

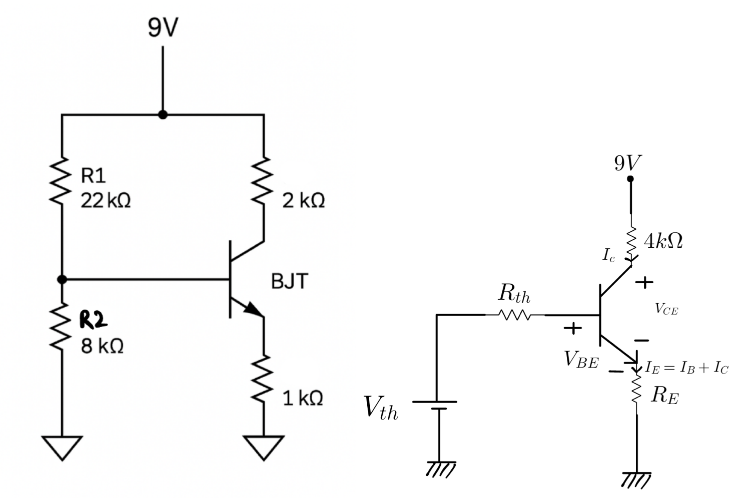

Fig. 2

$$V_{th} = \frac{R_2}{R_1 + R_2} \times V_{CC}$$
$$V_{th} = \frac{8K}{8K + 22K} \times 9V = \frac{8}{30} \times 9V$$

$$V_{th} = 2.4V$$
$$R_{th} = 8K||22K = 5.86k\Omega$$

KVL in Base-Emitter loop:

$$+V_{th} - I_B R_{th} - 0.8V - I_E R_E = 0$$

$$2.4V - I_B \times 5.86k\Omega - 0.8V - (\beta + 1)I_B \times R_E = 0$$

$$\frac{2.4V - 0.8V}{101k\Omega + 5.86k\Omega} = I_B$$

$$\boxed{I_B = 14.97\mu A} \quad \text{or} \quad \boxed{I_B = 0.01497mA}$$

$$I_C = \beta I_B = 100 \times 14.97\mu A$$

$$\boxed{I_C = 1.497mA}$$

$$I_E = I_C + I_B = 1.497mA + 0.01497mA$$

$$\boxed{I_E = 1.51mA}$$

KVL in Collector-Emitter loop:

$$+9V - 2k\Omega \times 1.497mA - V_{CE} - 1.5mA \times 1k\Omega = 0$$

$$9V - 2.98V - 1.5V = V_{CE}$$

$$\boxed{V_{CE} = 4.51V}$$

Soln (b) Region of operation:

$$V_{CE} = V_{CB} + V_{BE}$$

$$4.51V = V_{CB} + 0.8V$$

$$V_{CB} = 4.51V - 0.8V$$

$$\boxed{V_{CB} = 3.71V}$$

$$V_{BE} > 0 \quad \text{(forward)}$$

$$V_{CB} > 0 \quad \text{(reverse bias)}$$

Active region of operation.

Ques 2. (a) A transistor amplifier circuit is shown in Fig. 3. Calculate the Q-point on the output characteristics

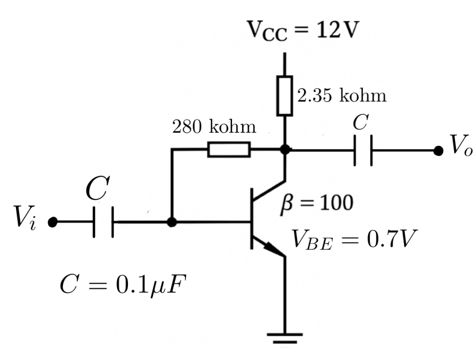

Fig. 3

Soln.

KVL in L1:

$$V_{cc} - R_c(I_C + I_B) - R_B I_B - V_{BE} = 0$$

$$I_B = \frac{V_{cc} - V_{BE}}{(\beta+1)R_c + R_B}$$

$$I_B = \frac{12 - 0.7}{2.35K\Omega \times 101 + 280K\Omega}$$

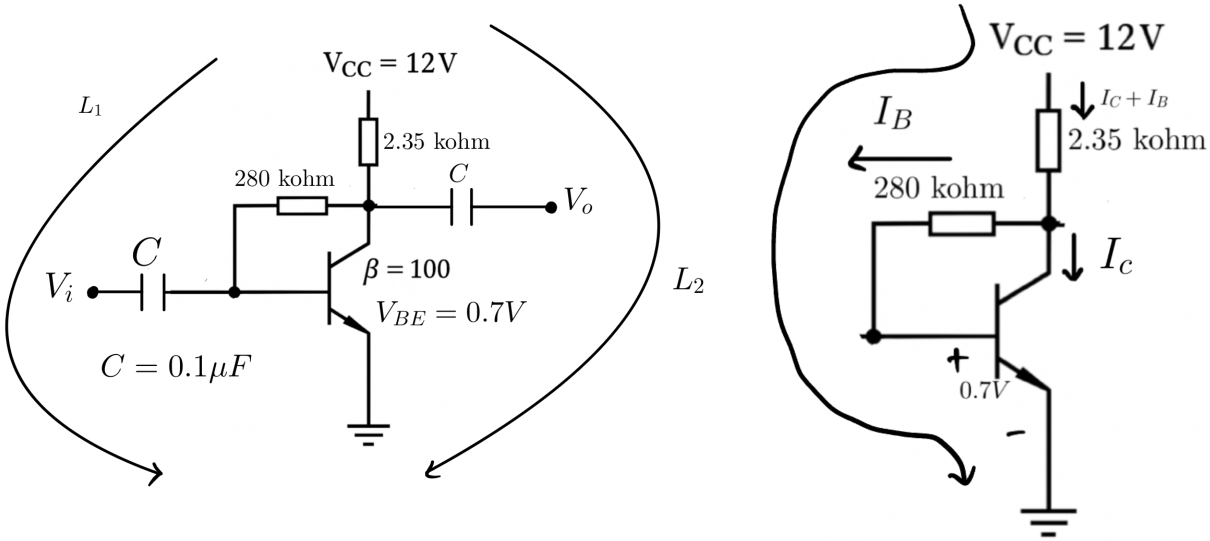

Fig. 4

$$I_B = \frac{11.3}{237.35K\Omega + 280K\Omega}$$

$$I_B = \frac{11.3}{517.35K\Omega} = 0.0218mA$$

$$\boxed{I_B = 0.0218mA \quad \text{or} \quad 21.8\mu A}$$

$$\boxed{I_C = \beta I_B = 100 \times 21.8\mu A = 2.18mA}$$

KVL in L2:

$$12 - 2.35K\Omega \times 2.2mA - V_{CE} = 0$$

$$\boxed{V_{CE} = 6.83V}$$